

ASSIGNMENT

OS with Linux as Case Study

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Component 1 - Linux Command Assignment

1. What command will you use to create a file called `summary.txt` and few lines of text in it?

```
~/./../../../../workground git:>main cat > summary.txt
This is line 1 of summary.
This is line 2 of summary.
This is line 3 of summary.
^C
~/./../../../../workground git:>main cat summary.txt
This is line 1 of summary.
This is line 2 of summary.
This is line 3 of summary.

~/./../../../../workground git:>main |
```

2. Create a folder named `backup`, and copy `summary.txt` into it without changing the filename.

```
~/./../../../../workground git:>main mkdir backup && cp summary.txt backup/
~/./../../../../workground git:>main ls -l backup/
total 4
-rw-r--r--. 1 user user 81 Aug  9 10:33 summary.txt

~/./../../../../workground git:>main |
```

3. Rename the file inside `backup/` from `summary.txt` to `summary_backup.txt`.

```
~/./../../../../workground git:>main mv backup/summary.txt backup/
summary_backup.txt
~/./../../../../workground git:>main ls -l backup/
total 4
-rw-r--r--. 1 user user 81 Aug  9 10:33 summary_backup.txt

~/./../../../../workground git:>main |
```

4. Create three files – *day1.txt*, *day2.txt*, and *day3.txt* – with one command.

```
~/./../../../../workground git:>main touch day1.txt day2.txt day3.txt
~/./../../../../workground git:>main ls -l day1.txt day2.txt day3.txt
-rw-r--r--. 1 user user 0 Aug  9 10:33 day1.txt
-rw-r--r--. 1 user user 0 Aug  9 10:33 day2.txt
-rw-r--r--. 1 user user 0 Aug  9 10:33 day3.txt

~/./../../../../workground git:>main |
```

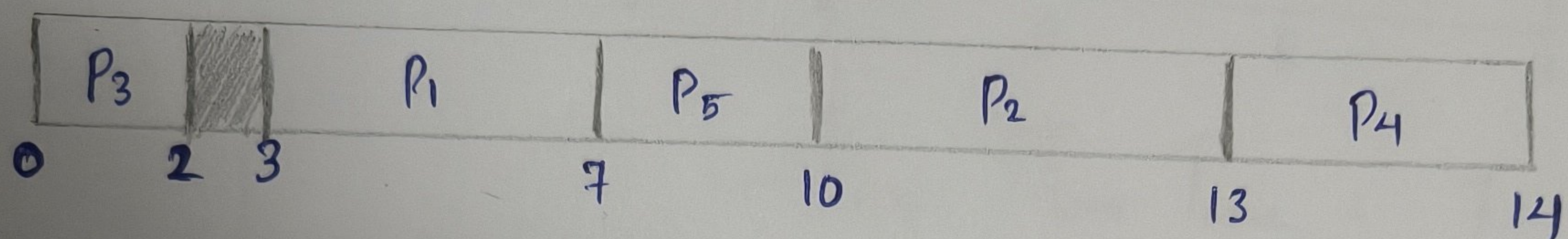
5. How many lines, words, and characters are in *summary.txt*?

```
~/./../../../../workground git:>main wc summary.txt
3 18 81 summary.txt

~/./../../../../workground git:>main |
```

7.)

Process	A.T	B.T	Exit Time	T.T	W.T	Response Time
P ₁	3	4	7	4	0	0
P ₂	5	3	13	8	5	5
P ₃	0	2	2	2	0	0
P ₄	5	1	14	9	8	8
P ₅	4	3	10	6	3	3

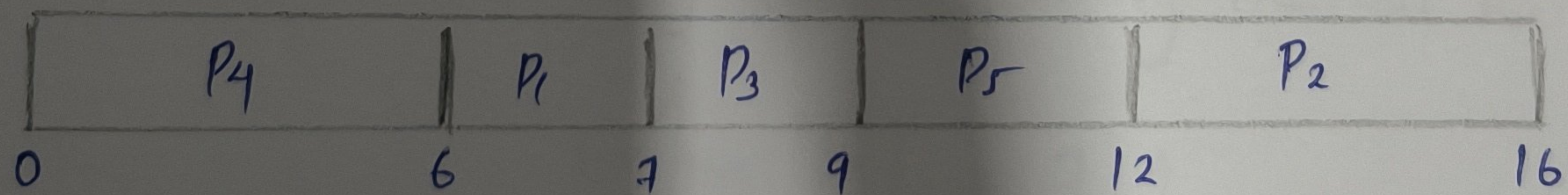


$$\text{Avg. Turn Time} : \frac{(4 + 8 + 2 + 9 + 6)}{5} = \underline{\underline{5.8}}$$

$$\text{Avg. Waiting Time} : \frac{5 + 8 + 3}{5} = \underline{\underline{3.2}}$$

8.)

PID	A.T	B.T	Exit Time	T.T	W.T	Response Time
P ₁	3	1	7	4	3	3
P ₂	1	4	16	15	11	11
P ₃	4	2	9	5	3	3
P ₄	0	6	6	6	0	0
P ₅	2	3	12	10	7	7



$$\text{Avg. Turn Time} : 40/5 = 8$$

$$\text{Avg. Waiting Time} : 24/5 = 4.8$$

Process	AT	BT	ET	TT	WT	RT
P ₁	3	1	4	1	0	0
P ₂	1	4	6	5	1	0
P ₃	4	2	8	4	2	2
P ₄	0	6	16	16	10	0
P ₅	2	3	11	9	6	6

P_4	P_2	P_1	P_2	P_3	P_5	P_4	
0	1	3	4	6	8	11	16

$$\text{Avg. Turn-Time} = 7$$

$$\text{Avg. Waiting Time} = 3.8$$

$$\text{Avg. Response Time} = 1.6$$

10)

PID	AT	BT	Priority	ET	TT	WT
P ₁	0	4	2	4	4	0
P ₂	1	3	3	7	6	3
P ₃	2	1	4	8	6	5
P ₄	3	5	5	13	10	5
P ₅	4	2	5	15	11	9

Assuming: lower priority number has the higher priority

$$\text{Avg. T.T} = 7.4$$

$$\text{Avg. WT} = 4.4$$

$$\text{Avg. RT} = 4.4$$

11)

Process	AT	BT	Priority	ET	TT	WT	RT
P ₁	0	4	2	4	4	0	0
P ₂	1	3	3	7	6	3	3
P ₃	2	1	4	8	6	5	5
P ₄	3	5	5	13	10	5	8
P ₅	4	2	5	15	11	9	9

P_1	P_2	P_3	P_4	P_5	
0	4	7	8	10	15

$$\text{Avg. T.T} = 7.4$$

$$\text{Avg. Wt. time} = 4.4$$

12)

Process ID	A.T	BT	ET	TT	WT	RT
P ₁	0	5	13	13	8	0
P ₂	1	3	12	11	8	1
P ₃	2	1	5	3	2	2
P ₄	3	2	9	6	4	4
P ₅	4	3	14	10	6	5

P_1	P_2	P_3	P_1	P_4	P_5	P_2	P_1	P_5	
0	2	4	5	7	9	11	12	13	14

$$\text{Avg. Turn Time} = 8.6$$

$$\text{Avg. Waiting Time} = 5.6$$

$$\text{Avg. Response Time} = 2.4$$